



**ETRO**  
**ELECTRONICS &  
INFORMATICS**

## TASK 3: VIRTUALIZATION

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### OPERATING SYSTEMS AND SECURITY (OSSEC)

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## Introduction

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Virtualization allows users to run multiple O.S on a single host machine. This capability is particularly useful to use software across different environments without requiring multiple physical devices. This report examines the process of setting up virtualization on macOS, utilizing Parallels Desktop, and deploying Windows 10 and Ubuntu LTS virtual machines (VMs). It also investigates the impact of RAM allocation and CPU core distribution on virtual machine performance.

## 1. Install VM tool (Parallels Desktop)

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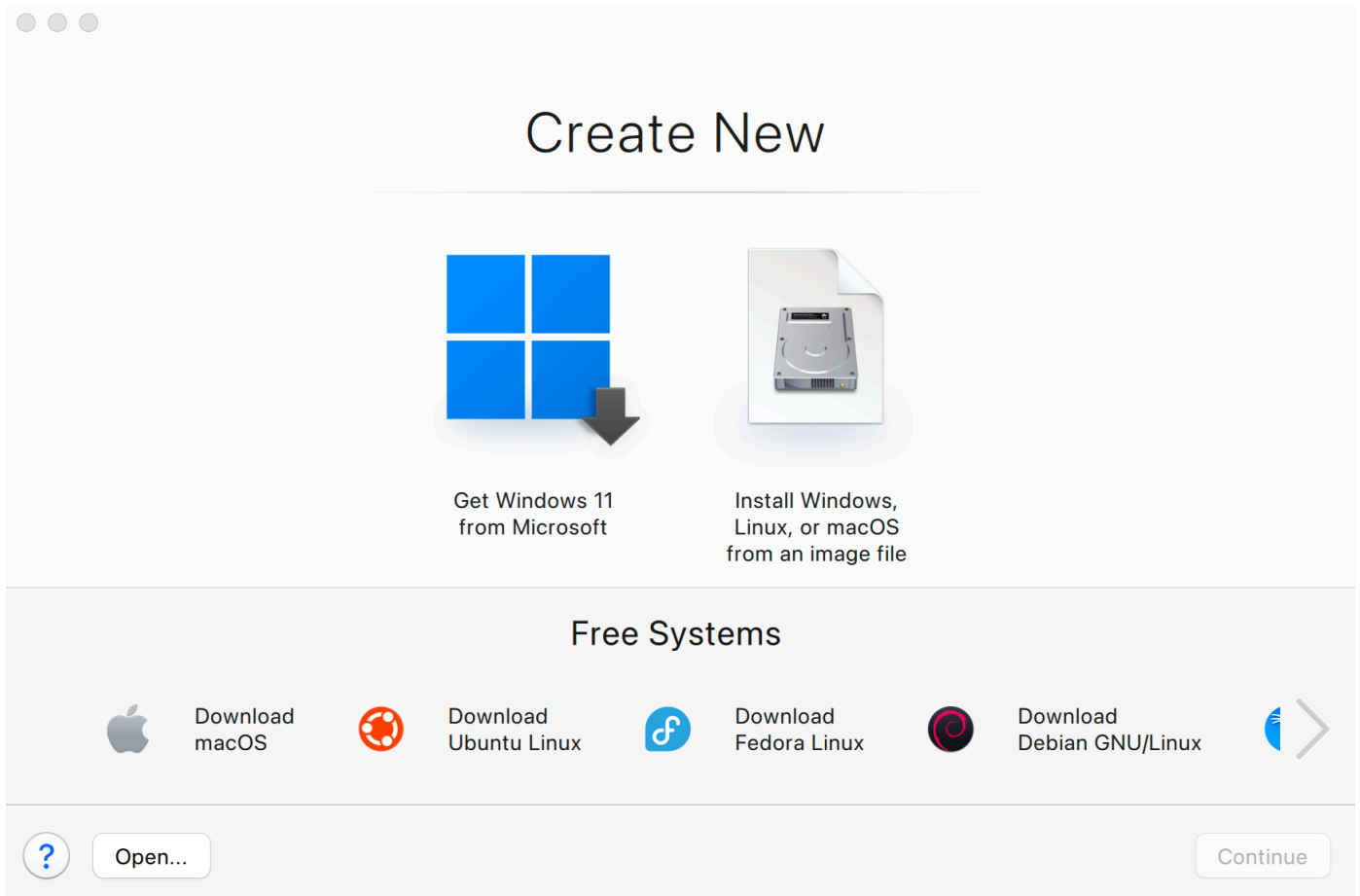
Since macOS does not natively support Microsoft Hyper-V or KVM, the most suitable virtualization tools for macOS include:

- Parallels Desktop
- VirtualBox

For this task, Parallels Desktop is used due to its optimized performance for macOS and integration with Apple Silicon and Intel-based Mac devices.

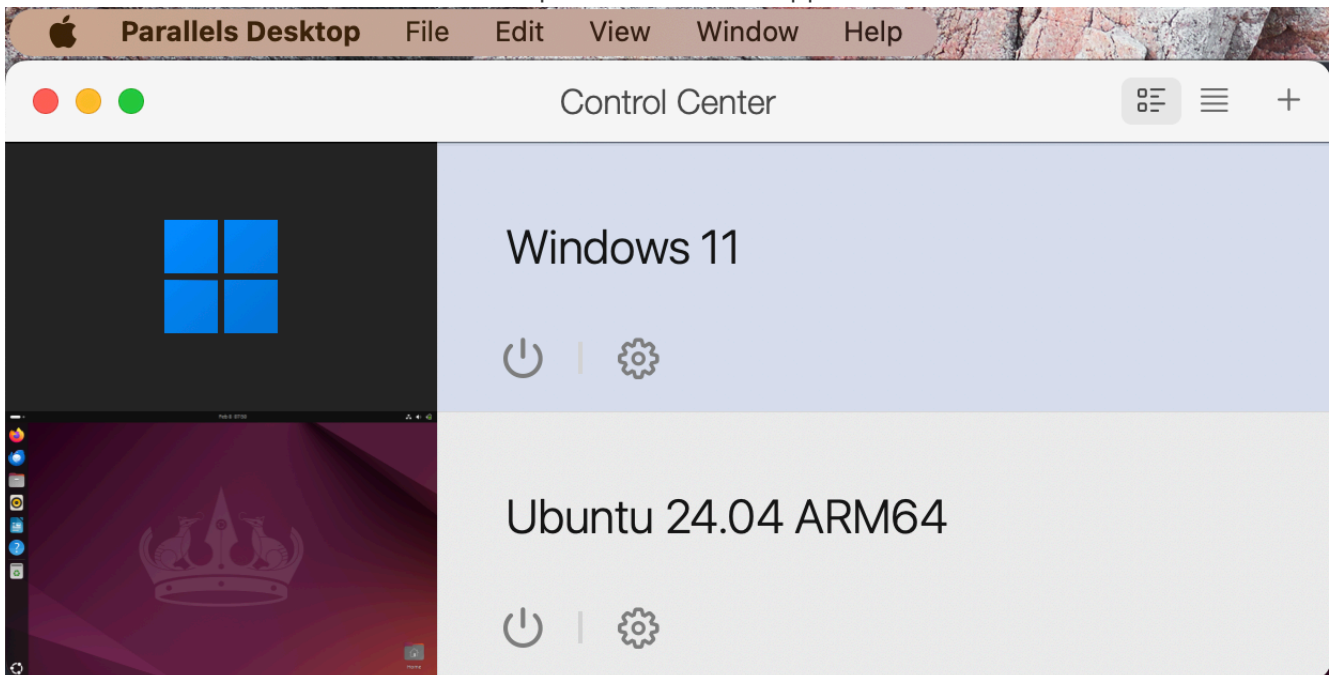
Installed Parallels Desktop via the below link:

<https://www.parallels.com/products/desktop/welcome/>



## 2. Create 2 VMs

- **Windows VM:** Used for compatibility with Microsoft applications.
- **Ubuntu VM:** Ideal for Linux-based development and server applications.



### 2.1. Motivation of RAM



The allocation of RAM in a VM is critical to ensure that the virtualized environments operate efficiently, preventing resource contention and maintaining optimal performance for both the VM and the host system.

General

Options

Hardware

Security

Backup

Q Search

! Some of the settings on this page cannot be changed until the virtual machine is shut down.

CPU & Memory

Graphics

Mouse & Keyboard

Shared Printers

Network

Sound & Camera

USB & Bluetooth

Hard Disk

Automatic (Recommended)  
For this Mac - 4 CPUs, 6 GB RAM (up to 3 GB for graphics)

Manual

Advanced...

Restore Defaults

Click the lock to prevent further changes.

?

General

Options

Hardware

Security

Backup

Q Search

! To be able to change the settings on this page, resume the virtual machine and then shut it down.

CPU & Memory

Graphics

Mouse & Keyboard

Shared Printers

Network

Processors: 2 (Recommended)

Memory: 2048 MB

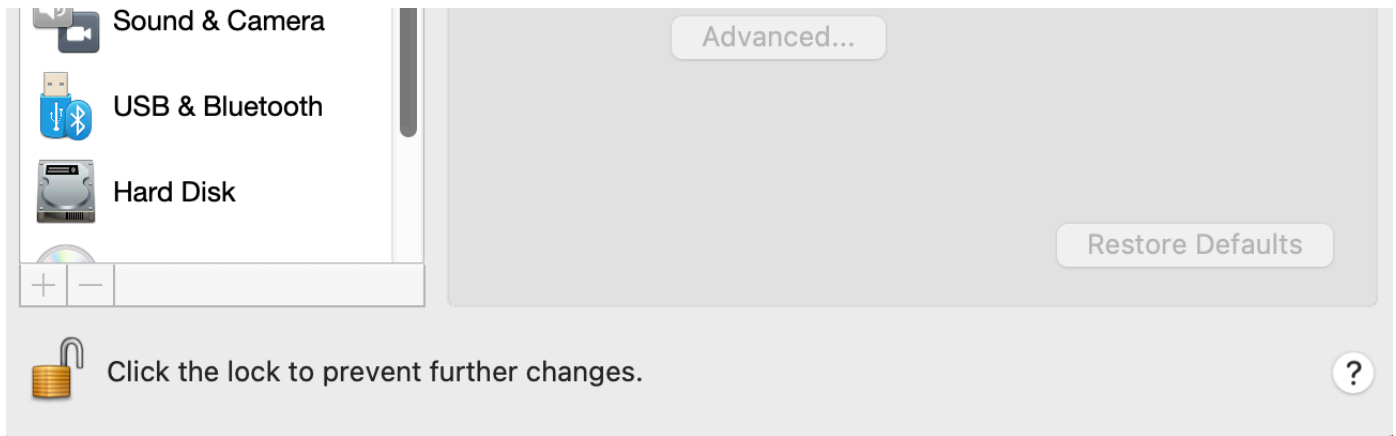
4 MB512 MB1 GB2 GB4 GB16 GB

Recommended

PRO

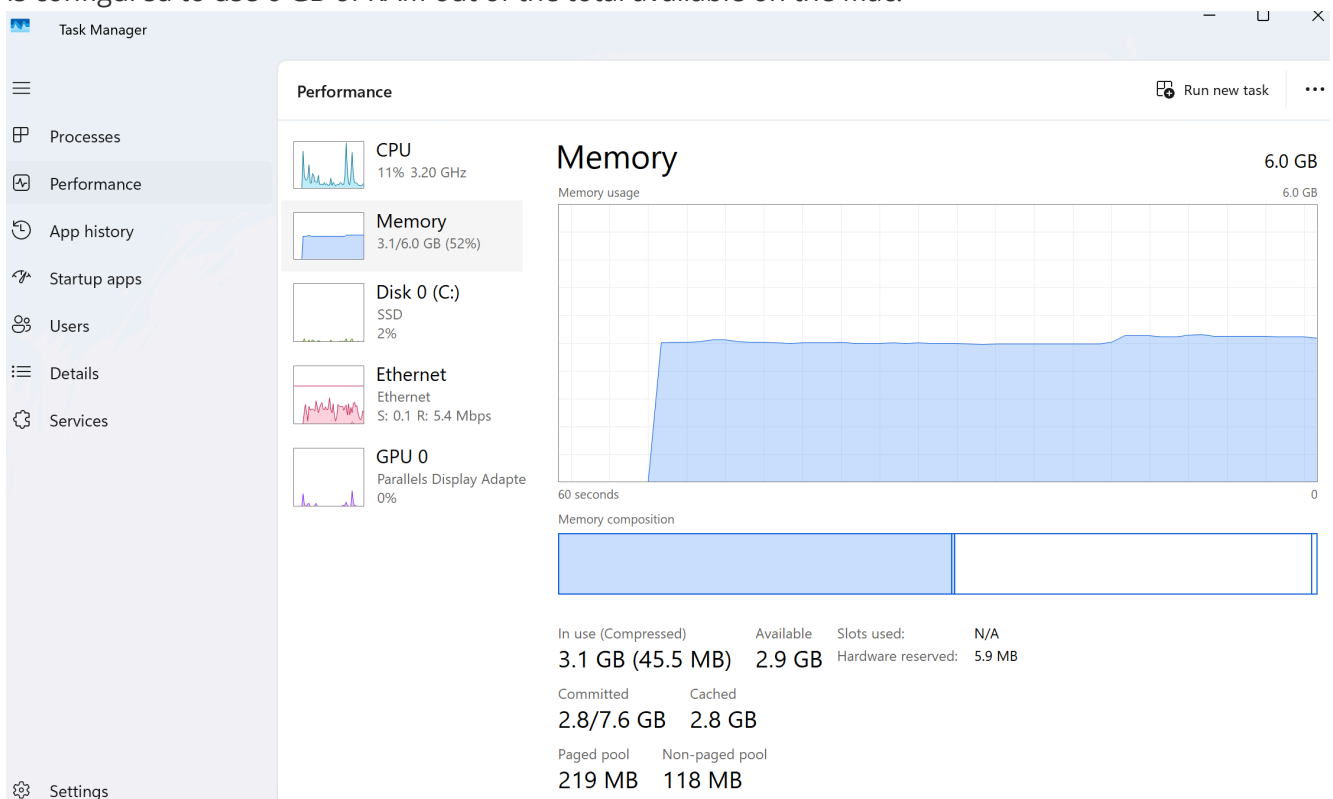
Use Rosetta to run x86-64 binaries

Performance may be affected



## RAM Allocation:

- Windows: The allocated RAM for the VM is displayed at the top right corner as 6 GB. This means the VM is configured to use 6 GB of RAM out of the total available on the Mac.



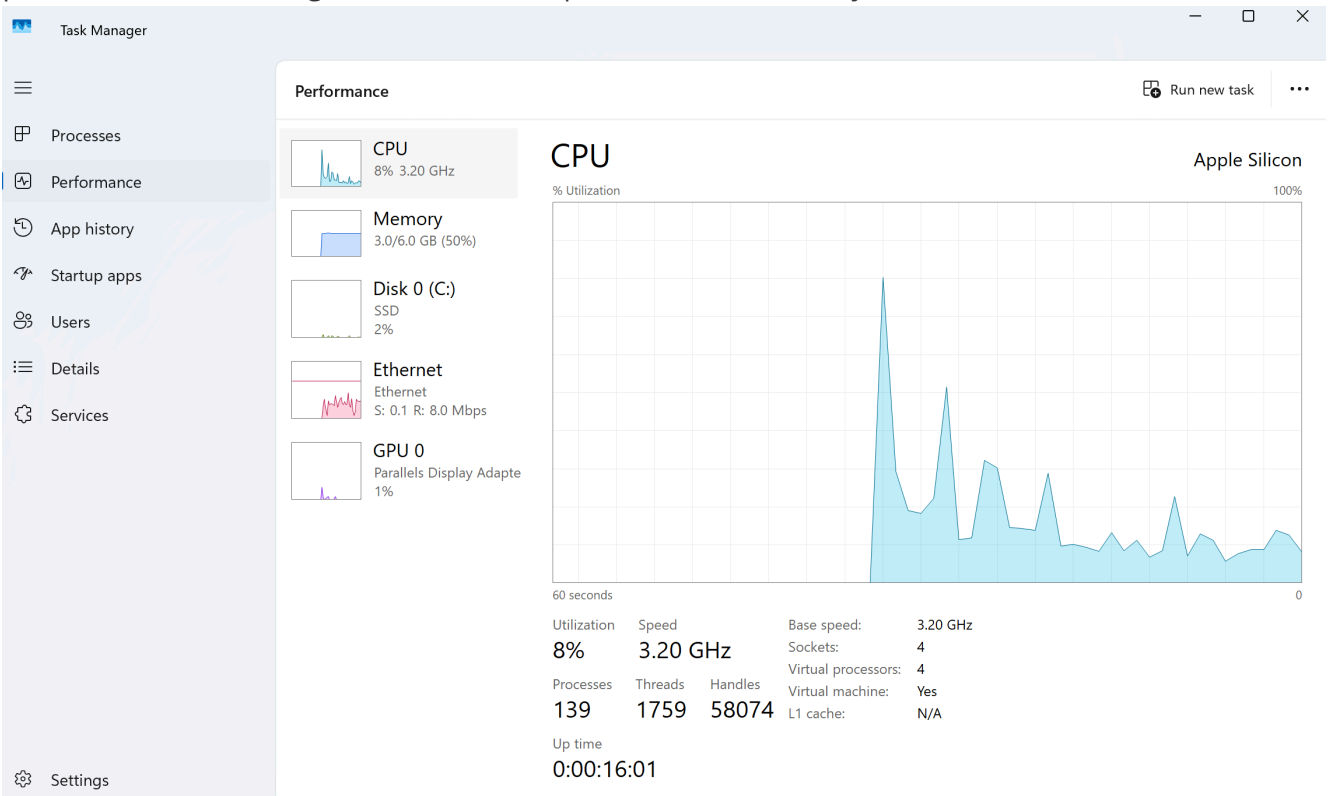
- Ubuntu: The ram is 2 GB.

```
parallels@ubuntu-linux-2404:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:          1.9Gi       835Mi       296Mi       117Mi       1.0Gi       1.1Gi
Swap:          2.0Gi           0B        2.0Gi
```

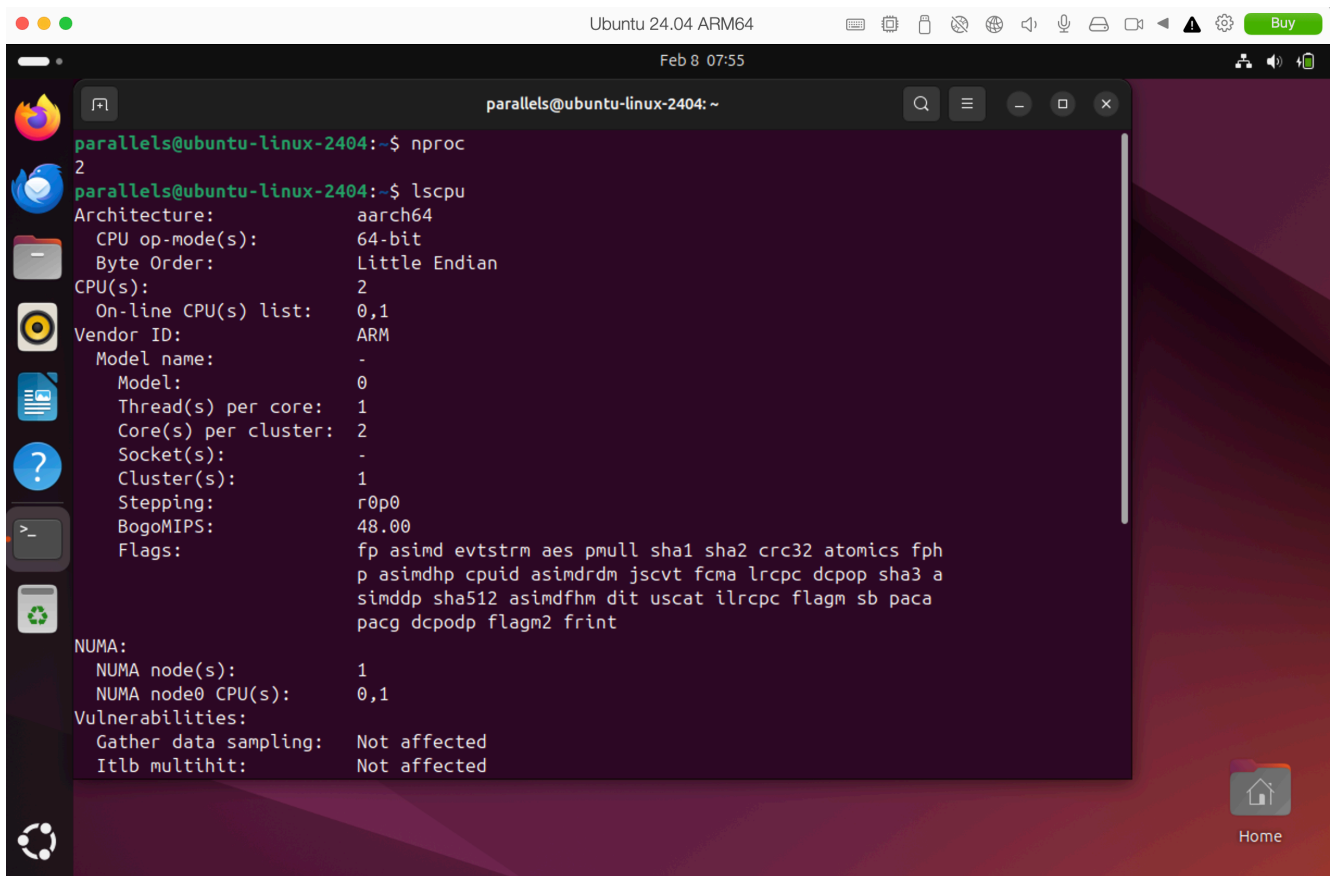
## 2.2. Motivation of number of cores

The number of CPU cores allocated to a VM affects its computational capabilities and overall performance. Allocating 2 cores per VM is generally sufficient, but increasing cores may enhance performance based on workload.

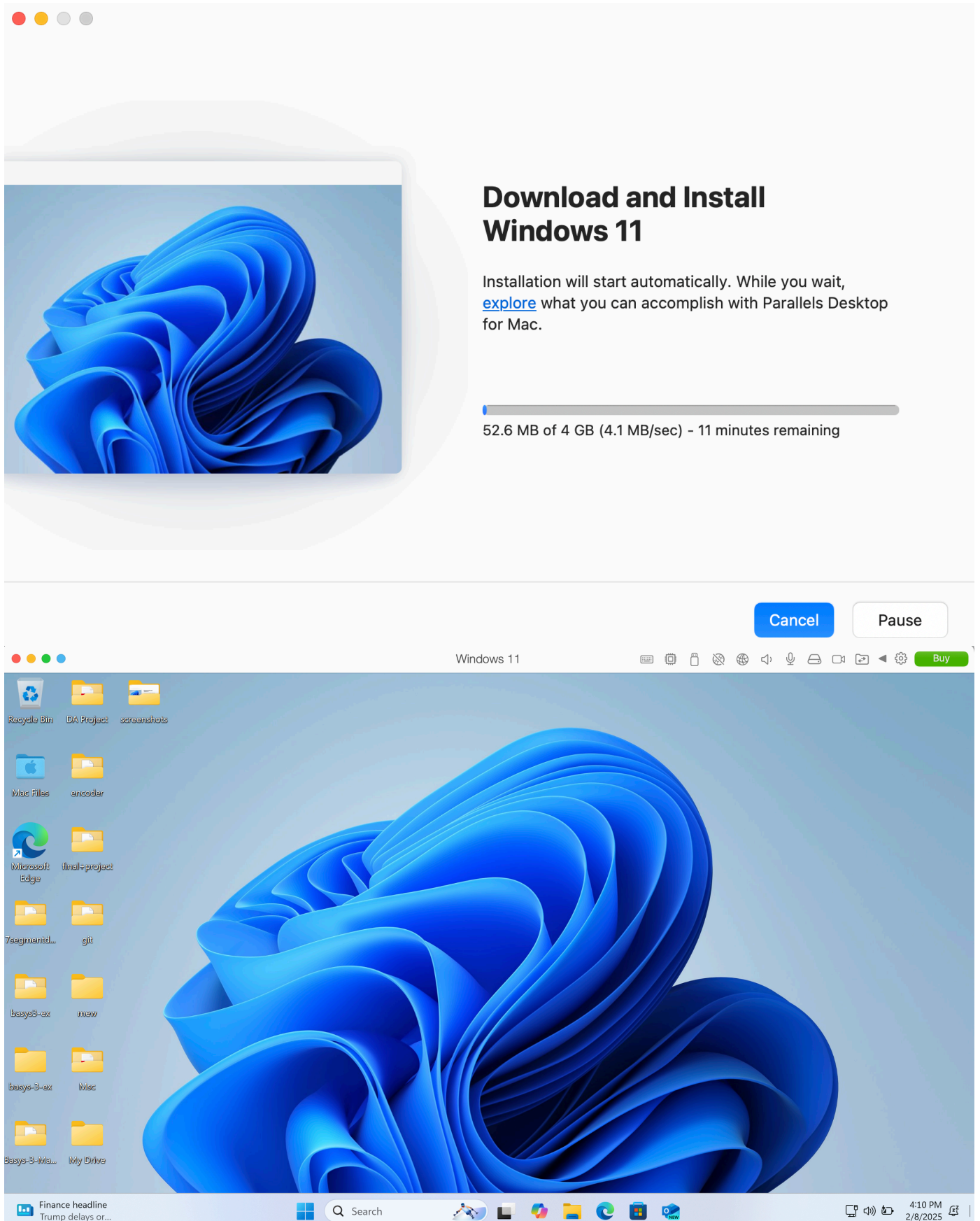
- **Windows:** the number of cores is represented by the label "Sockets" and the value of "Virtual processors". VM is configured with 4 virtual processors, which likely means 4 cores allocated.



-**Ubuntu:** the number of cores is 2.



### 3. Install Windows in one of the VMs

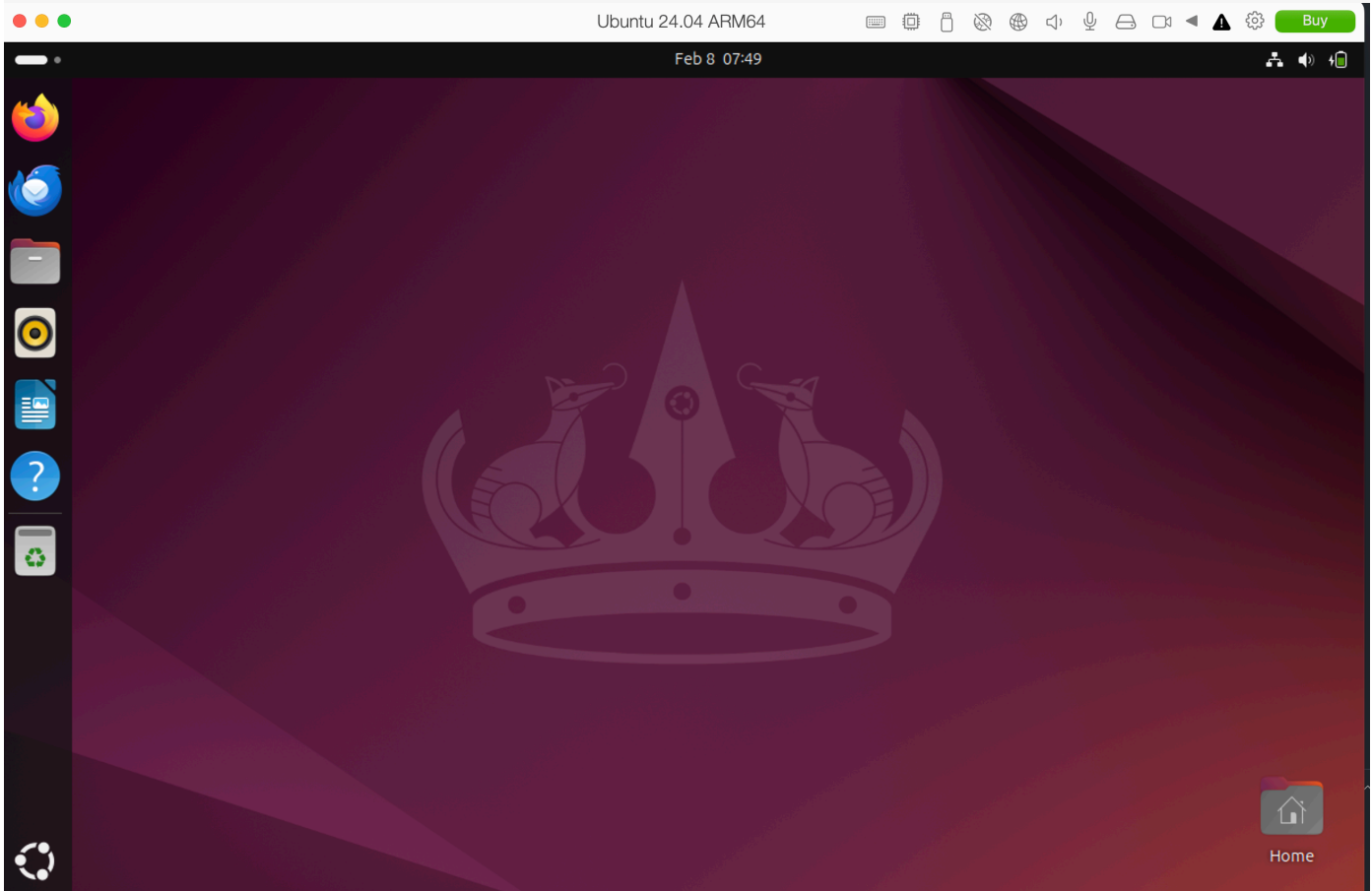


## 4. Install Ubuntu in the other VM

A screenshot of a terminal window showing the download progress of Ubuntu 24.04 ARM64. The window has a title bar with three colored buttons (red, yellow, grey) on the left. The main content area has a light grey background. At the top, the word "Downloading" is centered in a large, dark grey font. Below it, "Ubuntu 24.04 ARM64" is centered in a slightly smaller dark grey font. In the center is the Ubuntu logo, an orange circle with three white dots and curved lines. Below the logo, the text "Downloading..." is centered. At the bottom, a horizontal progress bar is shown, followed by the text "29.2 MB of 3.3 GB (5.3 MB/sec) - 13 minutes remaining" in a dark grey font.

Cancel

Pause



## Conclusion



Virtualization on macOS using Parallels Desktop enables seamless multi-OS deployment. By strategically allocating RAM and CPU cores, users can achieve optimal performance for both Windows and Linux environments.

## References

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- Apple Inc. "macOS Virtualization Overview." Apple Developer Documentation.
- Parallels Desktop Official Website. <https://www.parallels.com/>
- Microsoft. "Windows 10 System Requirements." Microsoft Documentation.
- Ubuntu. "Ubuntu LTS Installation Guide." Ubuntu Documentation.